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— 2026

: (SIFT), ( , ), () , RANSAC , .

:

[label=0.]

1. SIFT .
2. $k$-NN $(k = 2)$ $T \in [0.5, 0.95]$.
3. (cross-check).
4. .
5. RANSAC, $N$ OpenCV.
6. $\tau \in [1, 20]$ px .
7. 5 .
8. .

# 1

## 1.1 SIFT

SIFT (Scale-Invariant Feature Transform) - (DoG) 128- . $\mathbf{d}_1, \mathbf{d}_2$ :

$$D(\mathbf{d}_1, \mathbf{d}_2) = \|\mathbf{d}_1 - \mathbf{d}_2\|_2 = \sqrt{\sum_{i=1}^{128} (d_{1,i} - d_{2,i})^2} \tag{1}$$

(Lowe's ratio test): $m_1$ $m_2$ . , :

$$\frac{\|\mathbf{d} - \mathbf{d}_{m_1}\|}{\|\mathbf{d} - \mathbf{d}_{m_2}\|} < T, \quad T \in (0, 1) \tag{2}$$

$(T \approx 1)$, .

## 1.2

$H \in \mathbb{R}^{3\times 3}$ :

$$\mathbf{x}' \sim H\mathbf{x} \iff \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{3}$$

8 , 4 .

## 1.3 RANSAC

, (outliers) . RANSAC (Random Sample Consensus) :

1. $(s = 4)$.
2. $H$.

3. $M$ :

$$d_i = \|\mathbf{x}'_i - \hat{\mathbf{x}}'_i\|_2 = \left\| \mathbf{x}'_i - \frac{1}{\mathbf{h}_3^T \mathbf{x}_i} \begin{bmatrix} \mathbf{h}_1^T \mathbf{x}_i \\ \mathbf{h}_2^T \mathbf{x}_i \end{bmatrix} \right\|_2 \tag{4}$$

4. (), $d_i < \tau$.

5. , $H$ .

$N$, $p$ ($p = 0.99$):

$$N = \frac{\ln(1-p)}{\ln(1-(1-\varepsilon)^s)} \tag{5}$$

$\varepsilon = 1 - \frac{N_{\text{inliers}}}{N_{\text{total}}}$ — .

# 2

(boat1.jpg – boat5.jpg).

## 2.1 A.

:

- 1: 19 096 SIFT;
- 2: 12 804 SIFT.

cv2.BFMatcher(cv2.NORM_L2) $k$-NN ($k = 2$). $T$ (. ??).

1. $T$

| $T$ | 0.50 | 0.60 | 0.70 | 0.80 | 0.90 | 0.95 |
|---|---|---|---|---|---|---|
| | 2117 | 2854 | 3490 | 4268 | 6593 | 10298 |

$T = 0.75$ 3839 . (cross-check) 3485 . ( ' ).

. 1. (results/step5_raw_matches.jpg).
, .

## 2.2 B.

(method = 0):

$$H_{\text{LS}} = \begin{bmatrix} 9.337 \cdot 10^{-1} & -7.874 \cdot 10^{-2} & -8.721 \cdot 10^{2} \\ -1.917 \cdot 10^{-2} & 1.190 \cdot 10^{0} & -1.868 \cdot 10^{2} \\ -2.155 \cdot 10^{-5} & 4.409 \cdot 10^{-5} & 1.000 \cdot 10^{0} \end{bmatrix}$$

:

$$\text{Error}_{\text{LS}} = 59.64 \text{ px}$$

$\sim 12\%$ , .

. 2. (results/step7_warped_least_squares.jpg).
(Error = 59.64 px).

## 2.3 C. RANSAC

RANSAC $\tau = 3.0$ px :

- : 3060 3485 (87.8% );
- : $\varepsilon = 1 - 3060/3485 \approx 0.122$;
- $p = 0.99$:

$$N = \frac{\ln(1-0.99)}{\ln(1-(1-0.122)^4)} = \frac{-4.605}{\ln(1-0.596)} = \frac{-4.605}{-0.906} \approx 5.1$$

- : 0.496 px.

OpenCV (cv2.findHomography cv2.RANSAC):

- OpenCV RANSAC: 3050 , 0.537 px;
- RANSAC: 3060 , 0.496 px.

, .

$\tau \in [1,20]$ px (. ??).

2. $\tau$

| $\tau$ (px) | | (px) |
|---|---|---|
| 1 | 2766 | 0.401 |
| 2 | 3004 | 0.477 |
| 3 | 3050 | 0.537 |
| 5 | 3083 | 0.526 |
| 8 | 3092 | 0.532 |
| 12 | 3101 | 0.558 |
| 20 | 3195 | 4.161 |

: $\tau \in [2,12]$ ( $\approx 0.5$ px). $\tau = 20$ px 4.161 px (+645%), . $\tau = 3.0$ px.

. 3. $\tau$ (results/step11_tau_analysis.png).
2–12 px $\tau = 20$ px .

. 4. RANSAC (results/step12_inliers_outliers.jpg).
; .

## 2.4 D. 5

$I_0, I_1, I_2, I_3, I_4$ $I_2$. :

$$H_{2\leftarrow 2} = I, \quad H_{1\leftarrow 2} = H_{1\to 2}, \quad H_{0\leftarrow 2} = H_{1\to 2} \cdot H_{0\to 1} \tag{6}$$

$$H_{3\leftarrow 2} = (H_{2\to 3})^{-1}, \quad H_{4\leftarrow 2} = (H_{2\to 3})^{-1} \cdot (H_{3\to 4})^{-1} \tag{7}$$

4 $T$ ' . (distanceTransform) .
:

- 1–2: 0.537 px;
- 2–3: 0.544 px;
- 3–4: 0.911 px;

- 4–5: 0.919 px.

: **0.728** px, ( ).

. 5. 5 (results/step15_panorama.jpg).

## 2.5 E.

( 1 5), 0%.

- : 117 ;
- RANSAC 6 ;
- , .

# 3

[label=3.]

1. SIFT- , $(T = 0.75)$ 60% .
2. : 59.64 px 0.496 px RANSAC.
3. RANSAC. OpenCV (3060 3050).
4. $\tau$ $\tau \in [2, 12]$ px $\tau = 3.0$ px.
5. 5 0.728 px.